

Software Engineering

How To Create Software Products Everyone Loves!

Module 8

UI/UX Fundamentals: Colors

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Colors

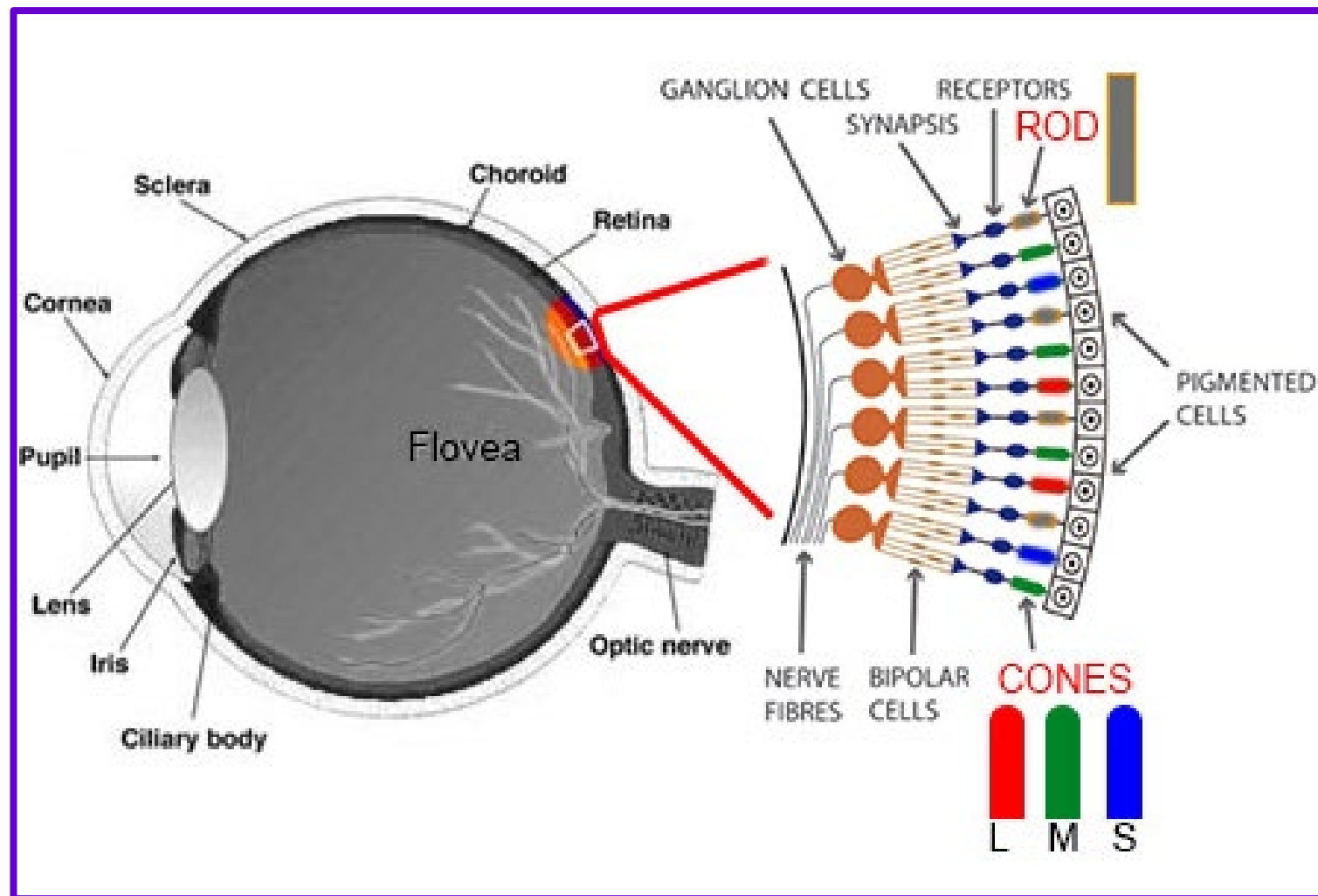
The hues we use affect the **style** and **mood** of the design, create the "**look & feel**" and are highly **memorable**.

Colors also have **strong ties with our emotions**, so we shouldn't choose them based on **aesthetics** alone.

According to many studies, over 90% of whether we like a product comes from the color palette that it uses.

ANATOMY OF COLOR

Color(**hue**) belongs in one of two groups:
colorful and **non-colorful** variants (white, greys and black).



Accessibility

Accessibility of a digital product is a science of making sure the product is suitable for older users, visually impaired, color-blind, and others

Color components like brightness and saturation influence the contrast, which directly affects the readability.

The higher the readability, the easier the product is to use for a **wider group** of people.

WCAG 2.0 Contrast Levels

Sufficient contrast levels help users with vision problems to be able to use the product as intended.

A

2.52:1

A - low contrast

AA

5.33:1

AA - average contrast

AAA

7.58:1

AAA - high contrast

Blue

COLOR PSYCHOLOGY

Calm, relaxation, trust, professionalism, experience, or wisdom.

WHEN TO USE BLUE?

IT, finance, banking, health, and social media.



Can you recognize the brand by its color?

From the left: Dropbox, PlayStation, Tumblr, Facebook, LinkedIn, Skype, Vimeo, Twitter

Green

COLOR PSYCHOLOGY

Health, nature, calm, and relaxation. harmony, stability, growth, security, or energy.

WHEN TO USE GREEN?

IT, finance and banking, health, fitness, ecology, and food.



Can you recognize the brand by its color?

From the left: Carlsberg, Starbucks, XBOX, Greenpeace, Evernote, Spotify, Android

Red

COLOR PSYCHOLOGY

energy, passion, strength, love, danger, adrenaline, warning, aggression

WHEN TO USE RED?

sports, food, services, cars and telecommunications.



Can you recognize the brand by its color?

From the left: KFC, RedBull, Toyota, Netflix, Rayban, CocaCola, Youtube

Yellow

COLOR PSYCHOLOGY

enthusiastic, self-confident, happy, optimistic, fun, warning

WHEN TO USE YELLOW?

food, sales, creative services, and art.



Can you recognize the brand by its color?

From the left: MailChimp, Ferrari, Kodak, McDonalds, Ikea, Snapchat, PostIt

Orange

COLOR PSYCHOLOGY

youth, energy, creativity, fascination, activity, bad taste or being cheap.

WHEN TO USE ORANGE?

food industry, sales, telecommunications, and products for children.



Can you recognize the brand by its color?

From the left: Harley-Davidson, Orange, Nike, Nickelodeon, Amazon, Lufthansa, McDonalds

Pink

COLOR PSYCHOLOGY

women, motherhood, innocence, youth, romance, gentleness, naivete and childishness

WHEN TO USE PINK?

cosmetics industry, fashion, pregnancy, female health, foundations, and non-profits



Can you recognize the brand by its color?

From the left: LG, Barbie, T-Mobile, Cosmopolitan, Johnson&Johnson, Airbnb, Victoria's Secret

Purple

COLOR PSYCHOLOGY

luxury, power, wealth, and secrecy, professionalism, wisdom, trust, high quality, and modernity

WHEN TO USE PURPLE?

IT industry, luxury goods, finance, banking, and food



Can you recognize the brand by its color?

From the left: Premier League, Twitch, FedEx, Yamaha, Yahoo!, Cadbury, Milka

Black and Grey

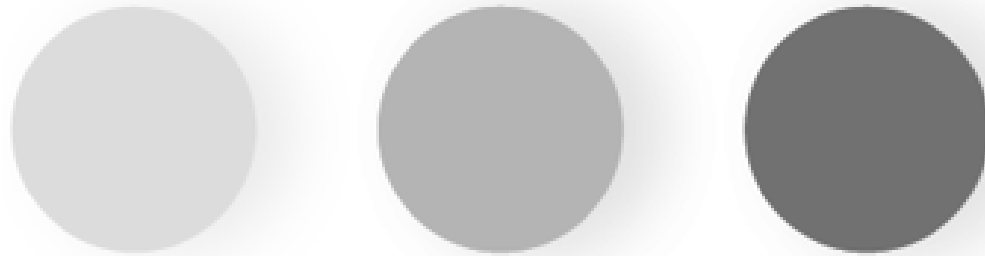
Black and shades of grey are serious, formal, and emotionally neutral colors.

They're associated with elegance, minimalism, professionalism, and luxury

They can, however, feel a bit exclusive and, when used in excess, can be quite depressing

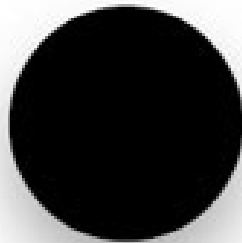
Grey

Light-grey items suggest being inactive or disabled, so try not to use it for essential objects or text.

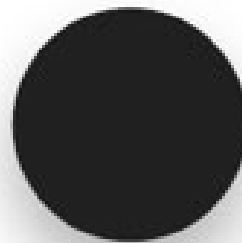


Black

Try to use very dark shades of grey instead of pure black whenever possible.



Pure black
(#000000)



Dark grey (#202020)

White

White is the color of minimalism, often suitable for creating clean, "airy" designs.

White is also the base for the name "white space," which is the negative space that sets hierarchy and order in design.

It's often associated with clarity and sterility. It's popular in both architecture, art, and fashion.

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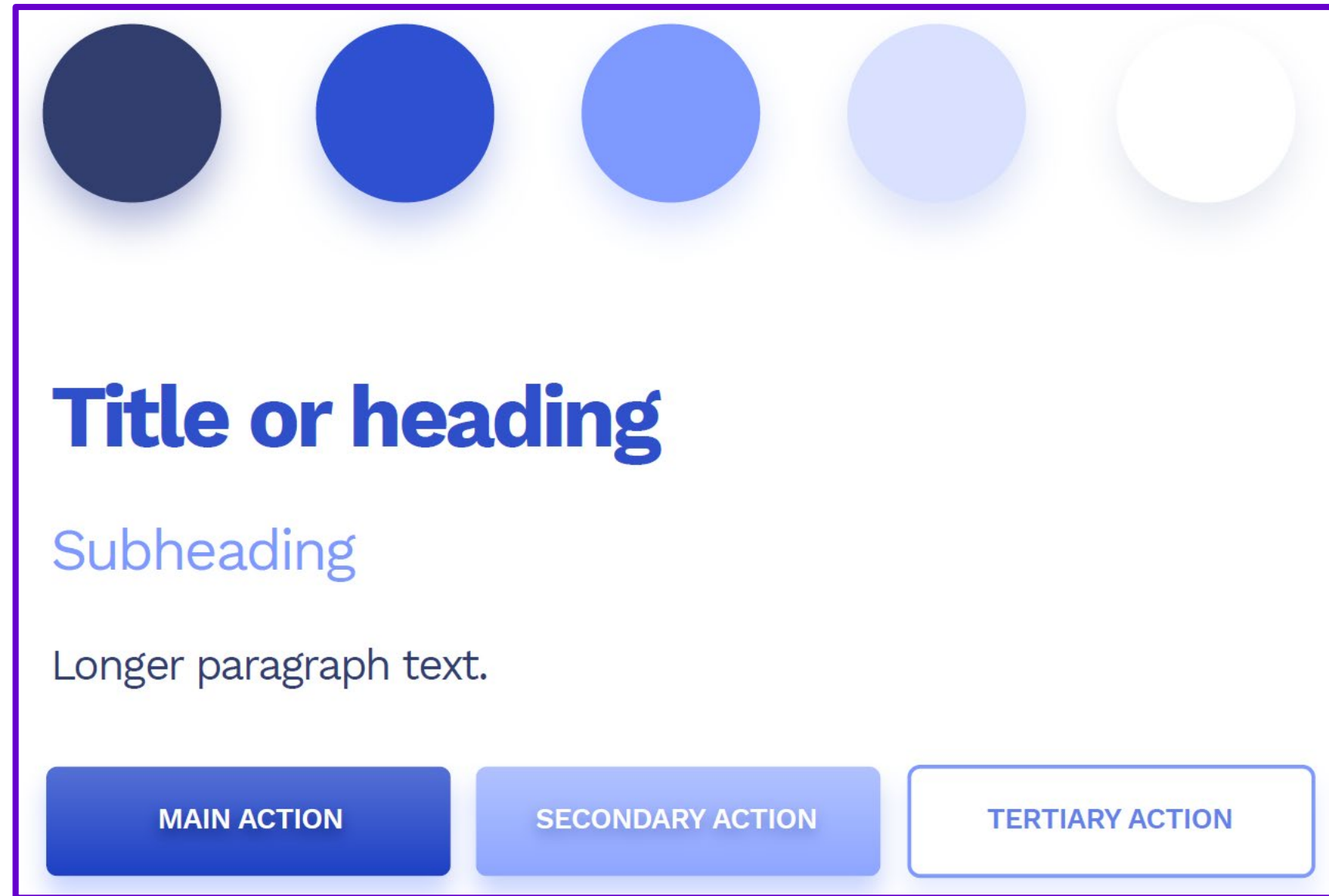
Gradient Best Practices

Color Palettes

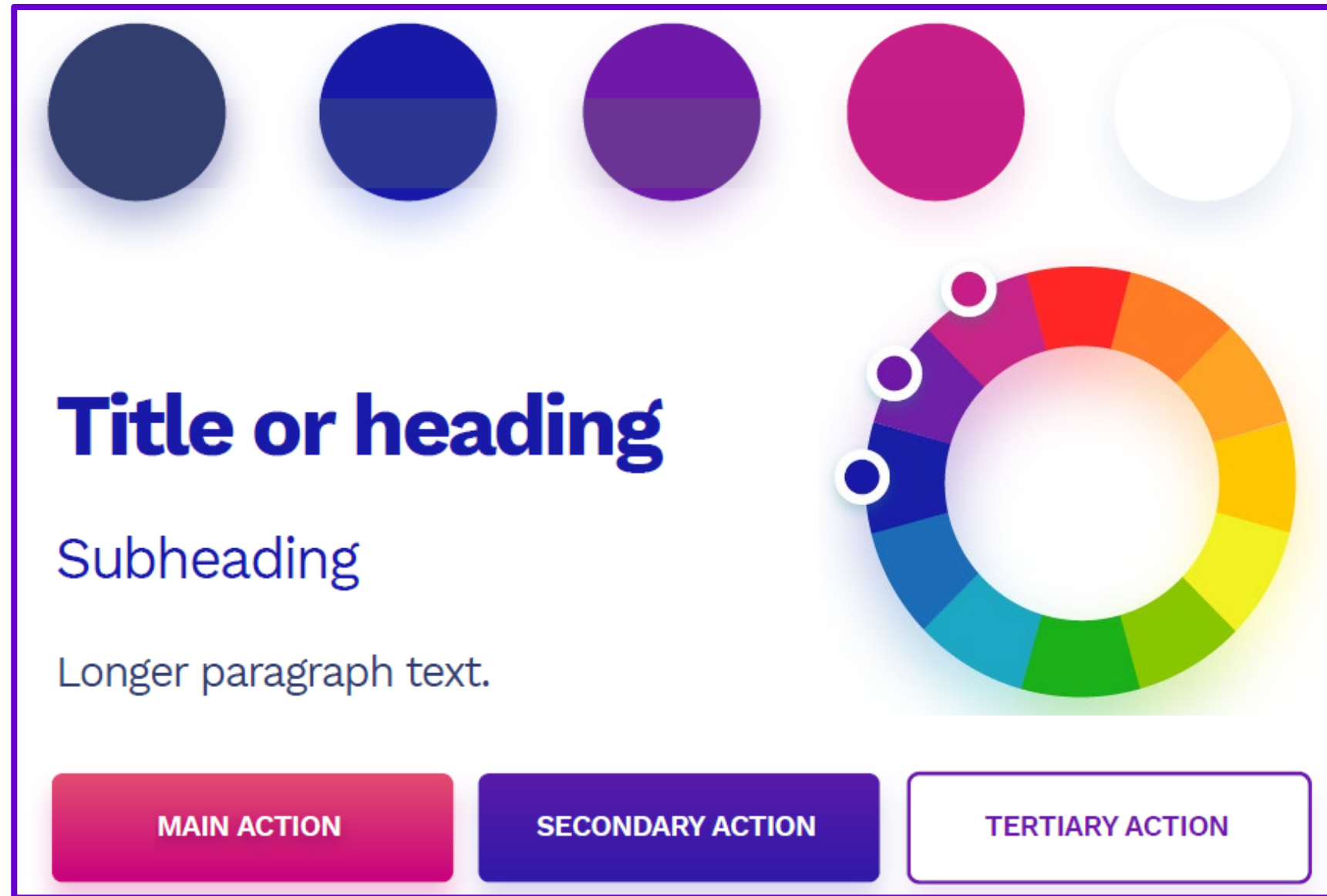
A color palette is a set of colors that work well together, combined to form a brand or a concept.



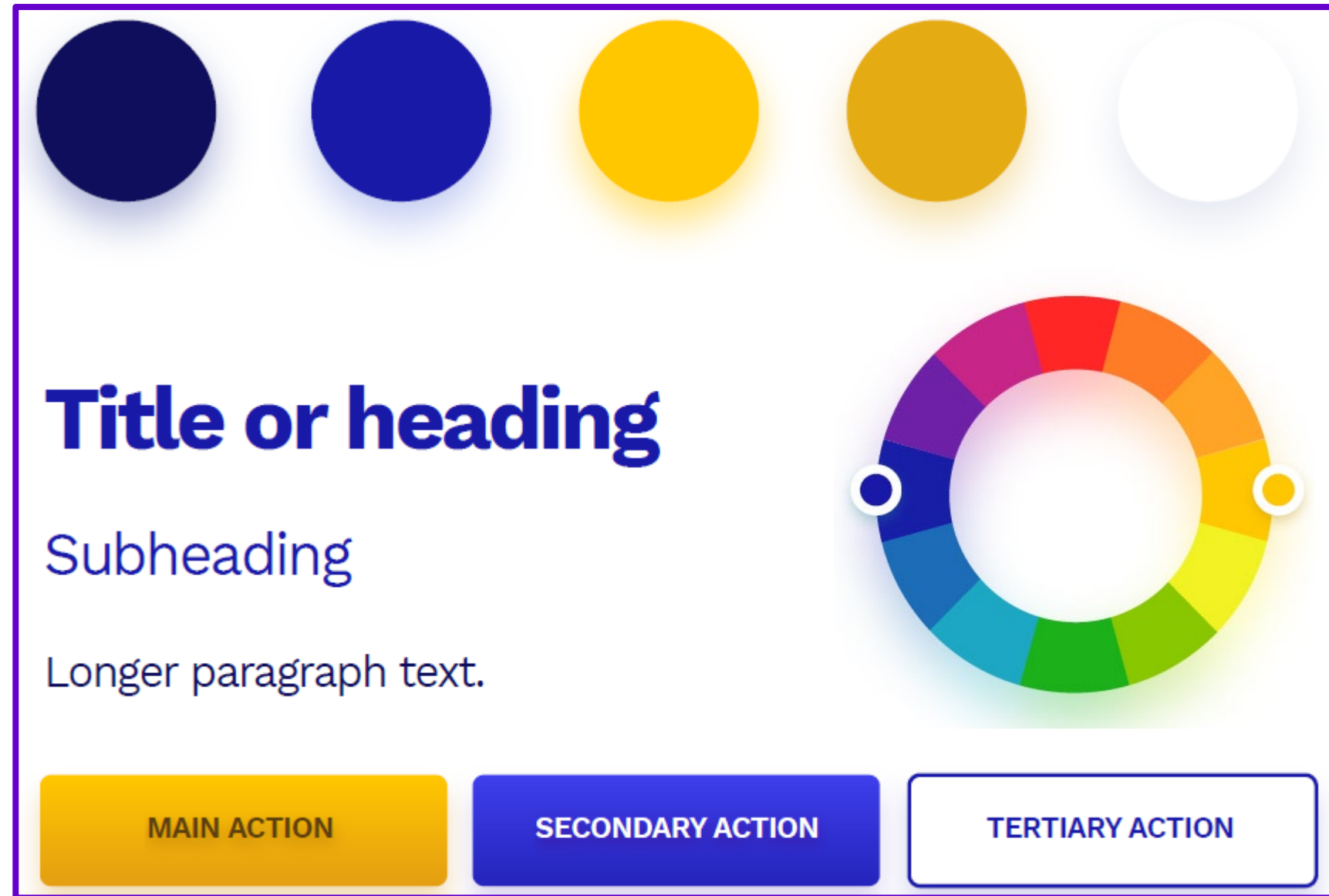
Monochromatic palette



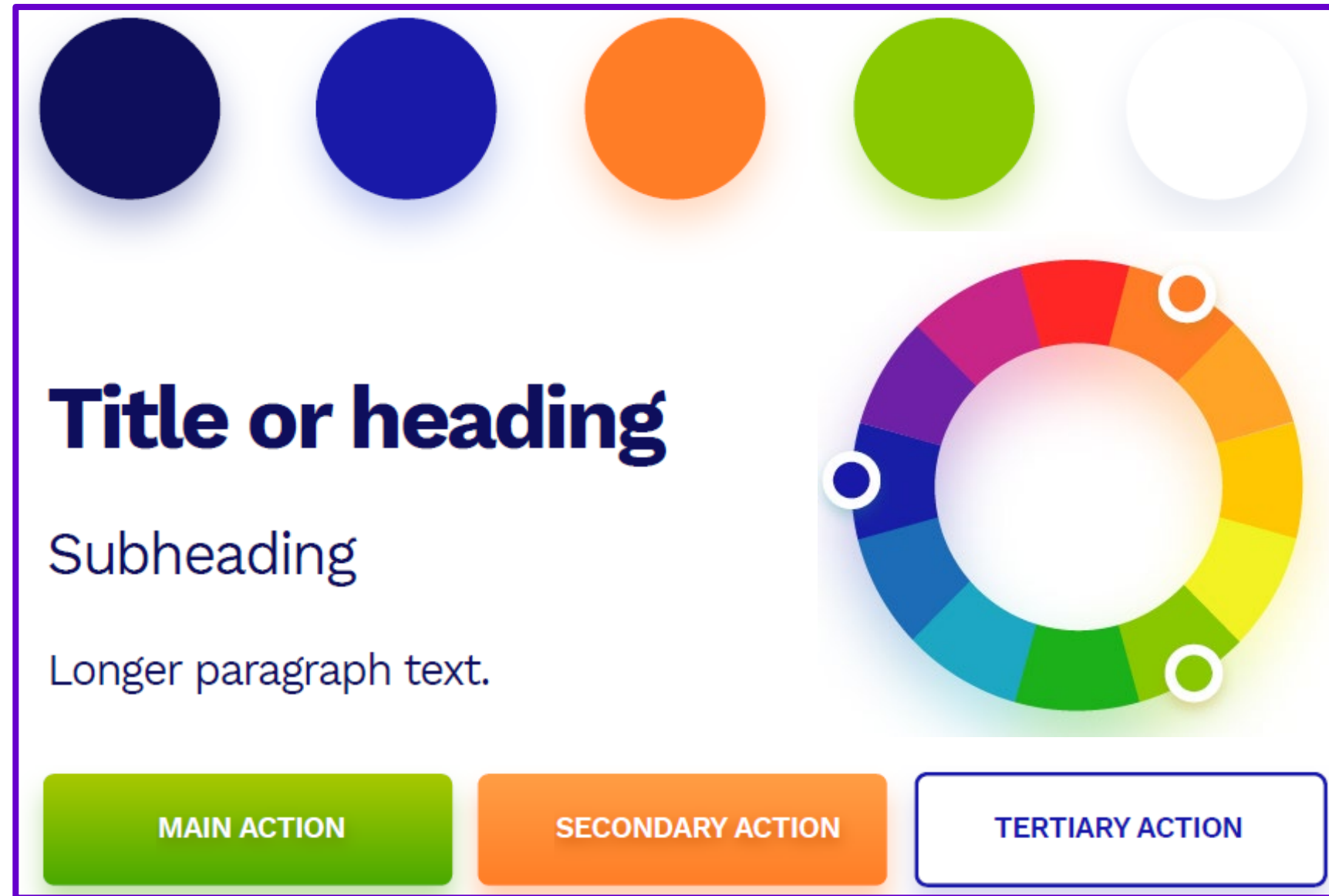
Analogous pallet



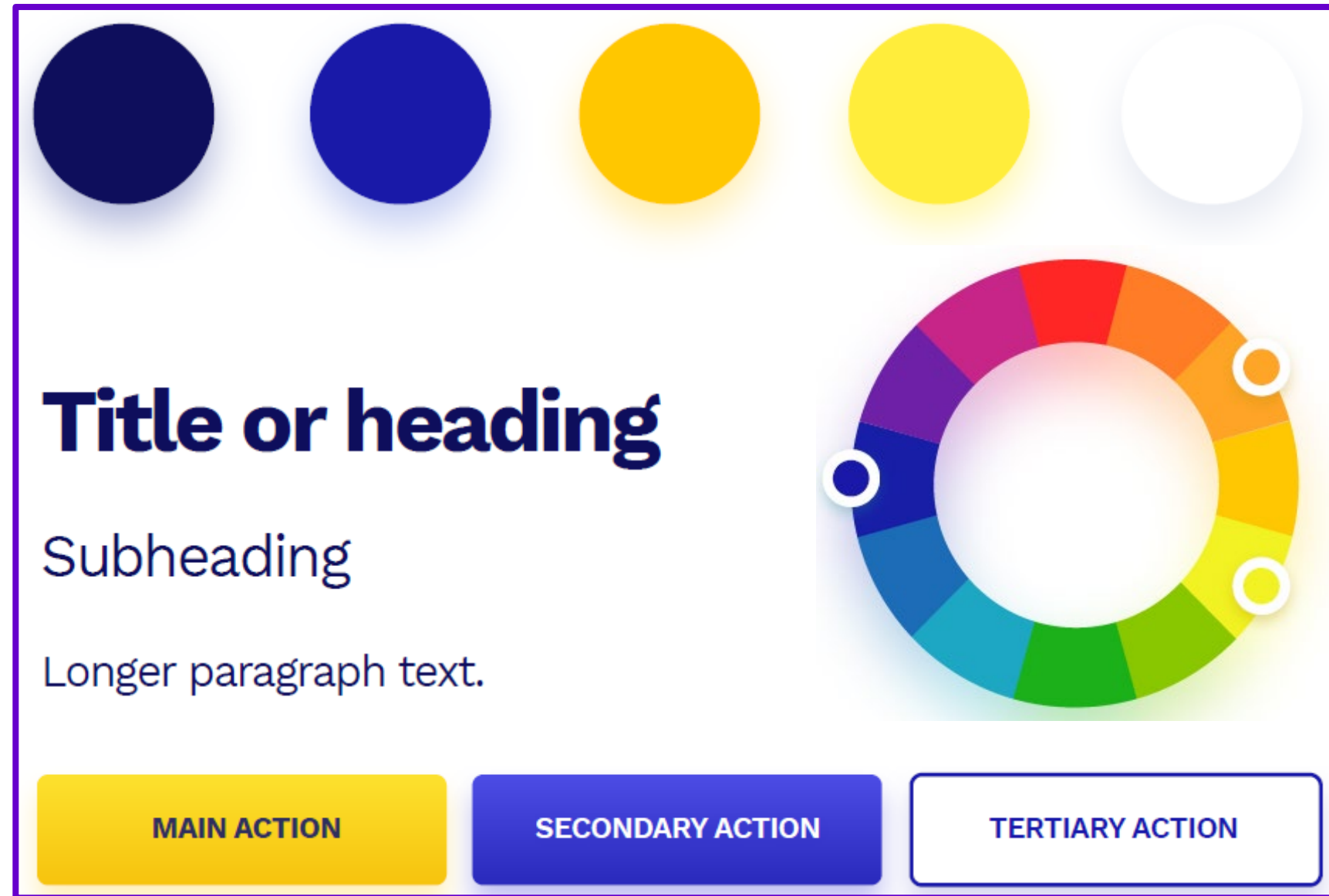
Complementary pallet



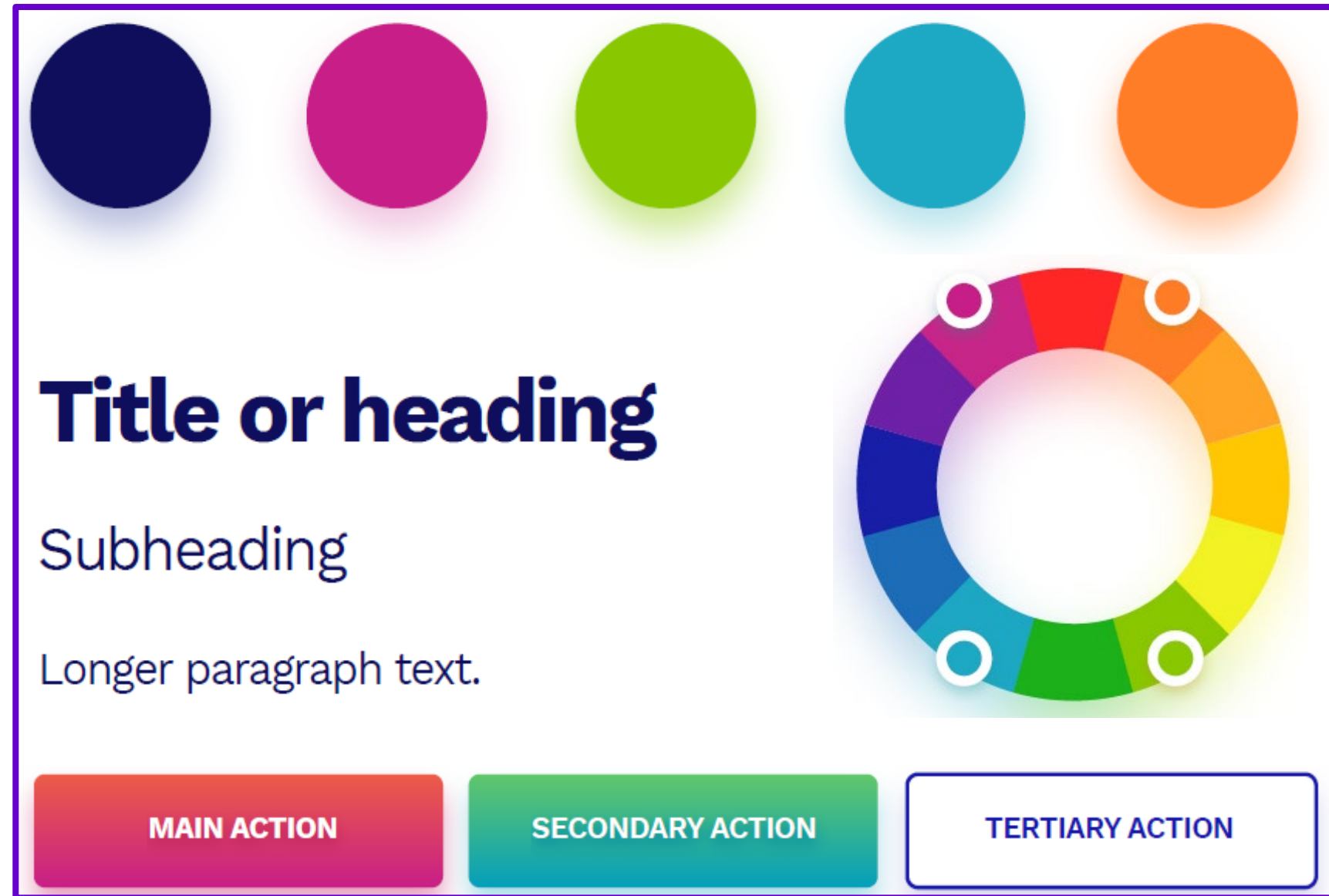
Triadic pallet



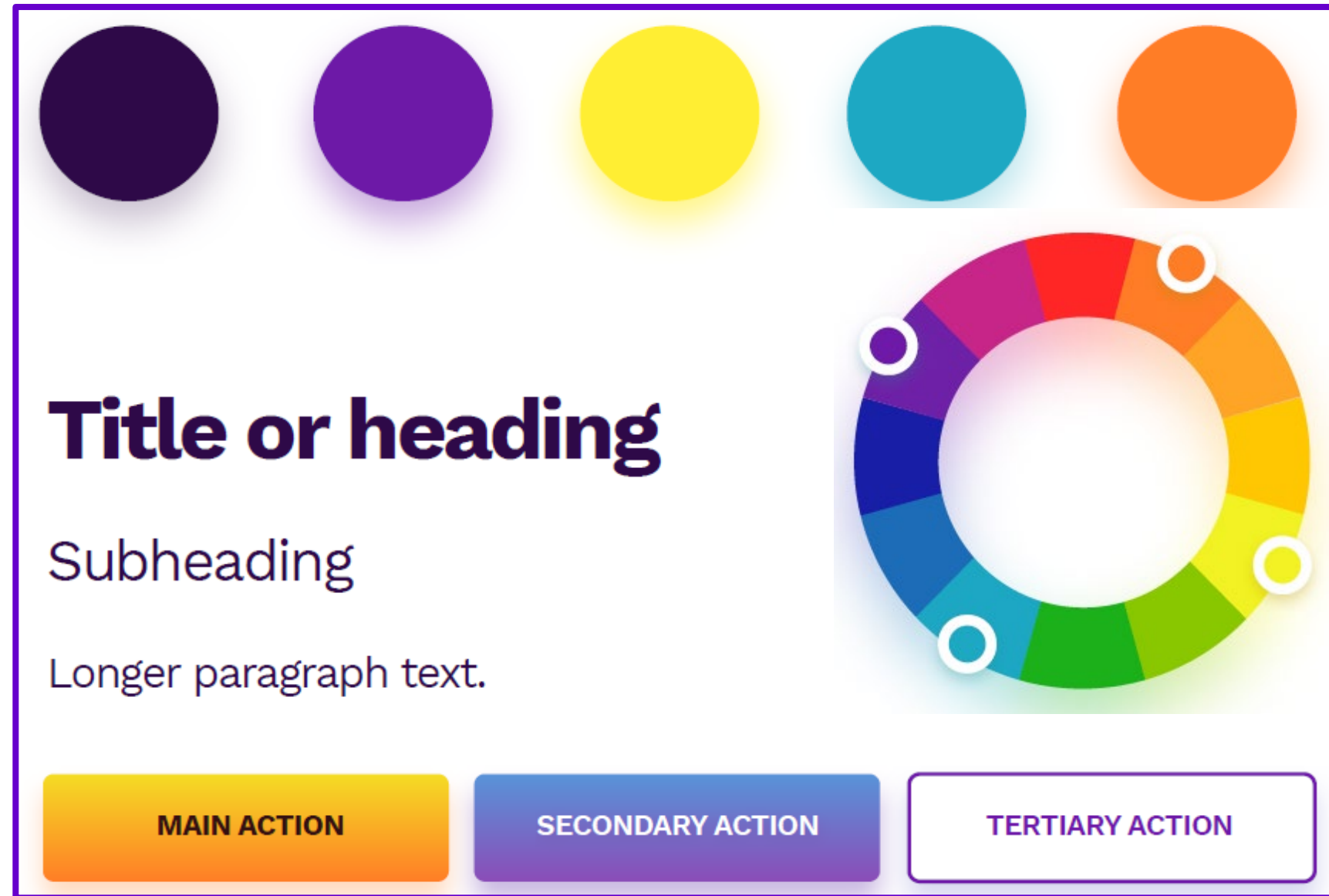
Split- complementary pallet



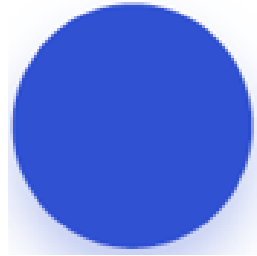
Rectangular pallet



Square pallet



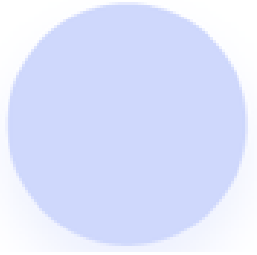
Greys



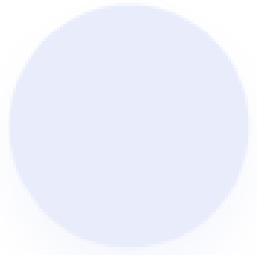
Primary color (#2F51D2)



Pick your primary color and decrease its Lightness and Saturation to about 30. You'll end up with a dark shade of grey mixed well with the primary hue.



To create the mid-range grey try setting the Brightness to 90 and decrease Saturation by ten from the initial value.



The lightest primary-grey shade uses even higher Brightness, at around 95. This shade works well as a background color.

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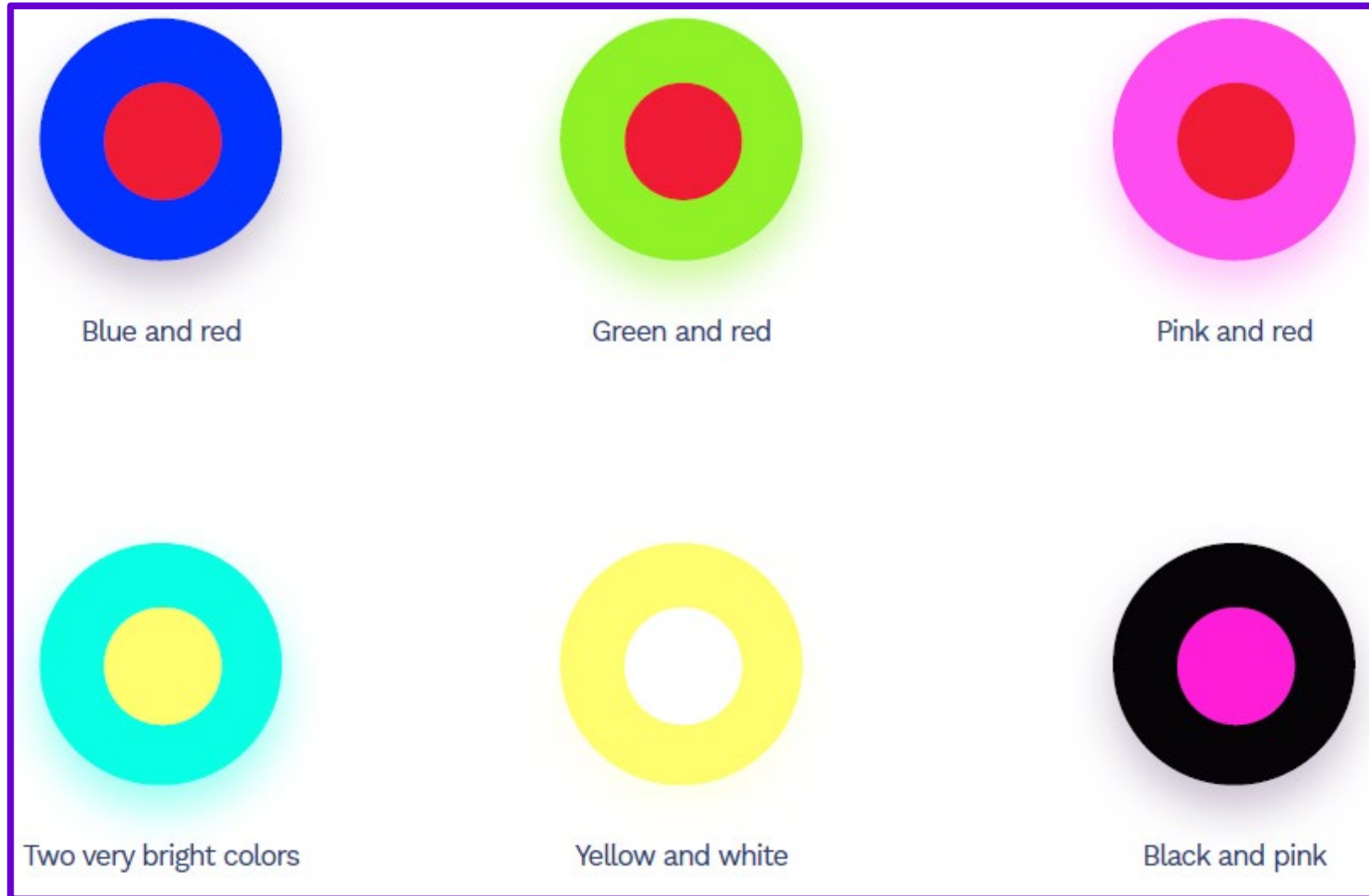
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Gradient Types

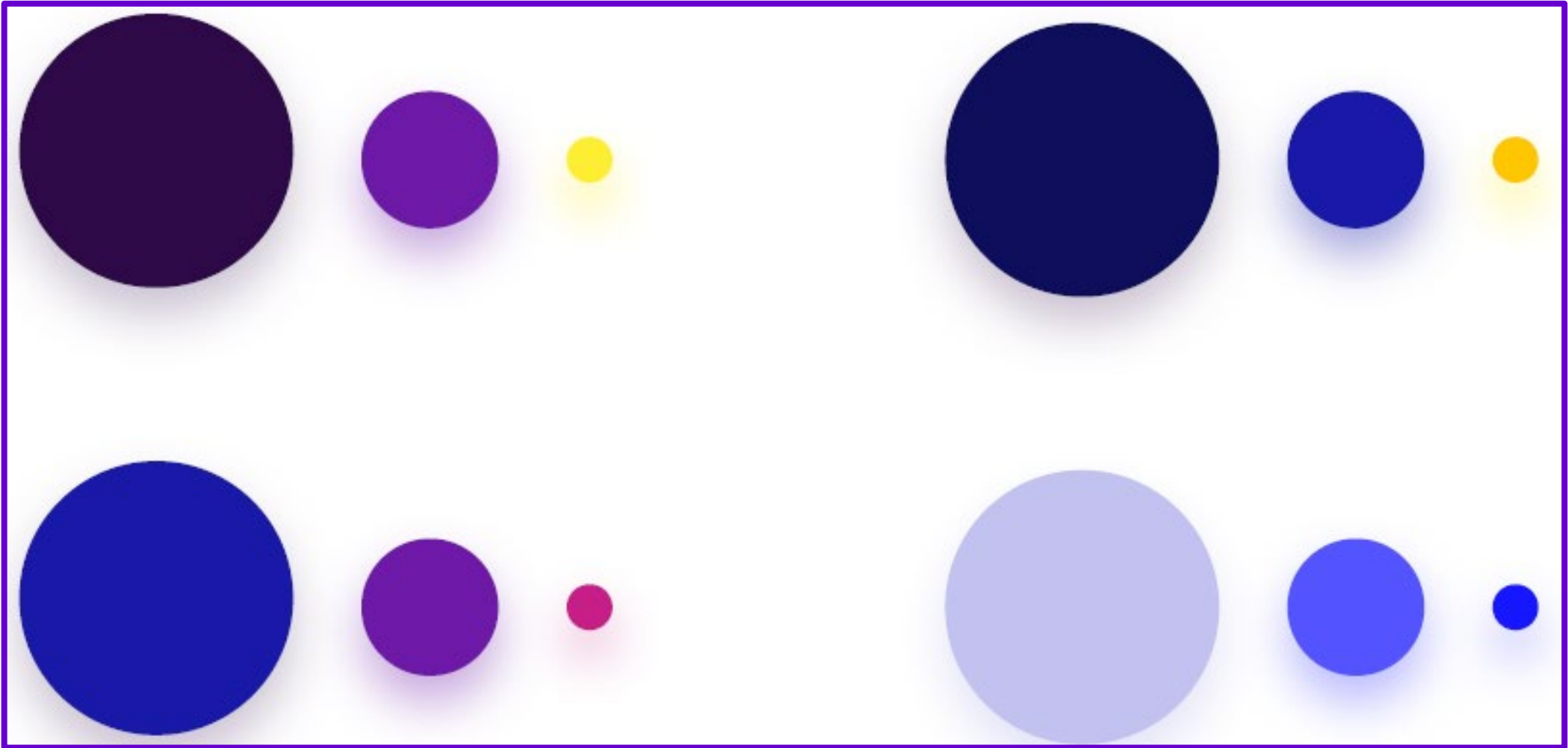
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Gradient Best Practices

COLORS THAT DON'T GO WELL TOGETHER



BALANCING A PALETTE (GOLDEN RATIO 60/30/10)

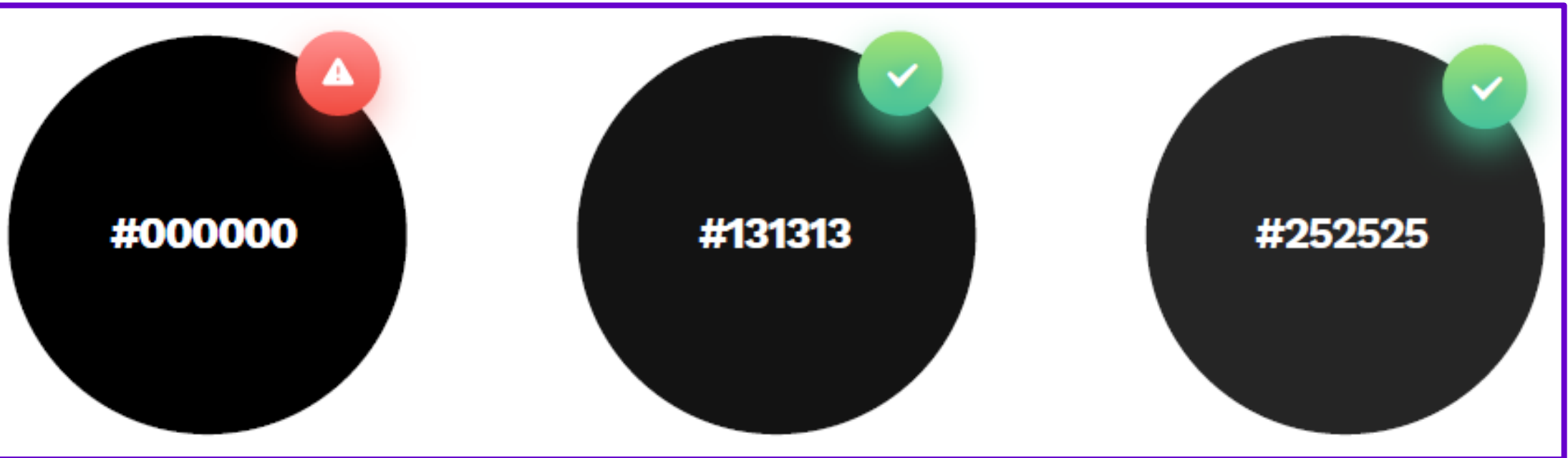


SATURATION



PURE BLACK COLOR DOESN'T EXIST

While pure black gives the best possible contrast (especially on a white background), it's not the most optimal choice for digital design.



SYSTEM STATES

Positive (success / confirmation) - green or blue

Payment successful.

Your subscription is confirmed.

SYSTEM STATES

Negative (error / failure) - red

Payment was declined by your bank.

SYSTEM STATES

Neutral - grey, blue

Settings saved.

Settings saved.

SYSTEM STATES

Warning - orange or yellow

Your subscription is expiring soon.

Terms and Conditions have changed.

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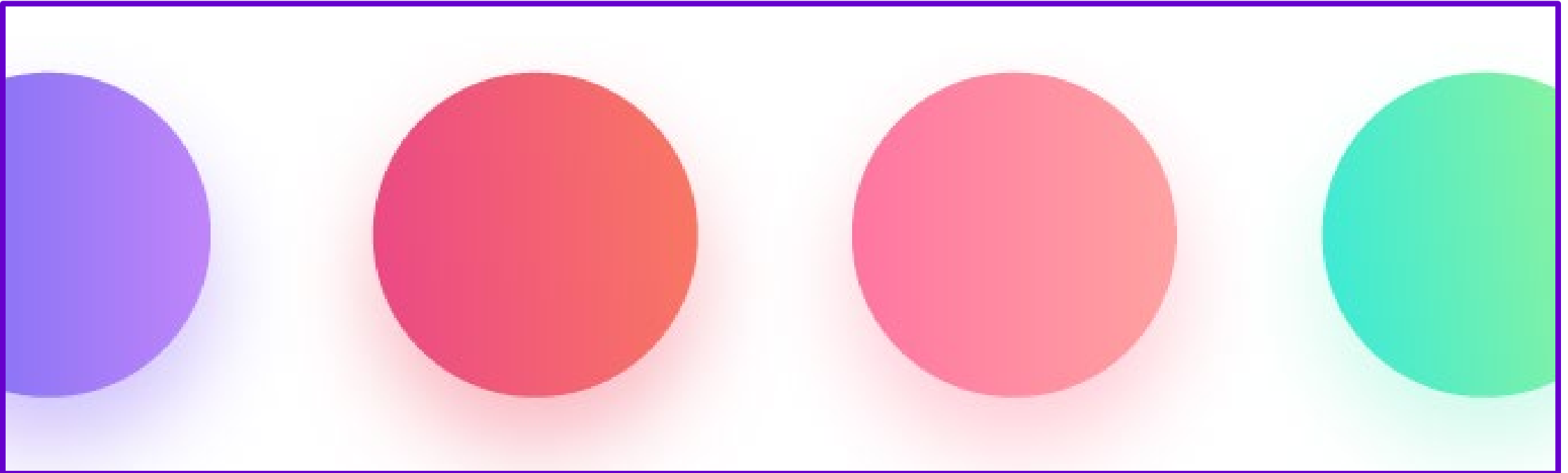
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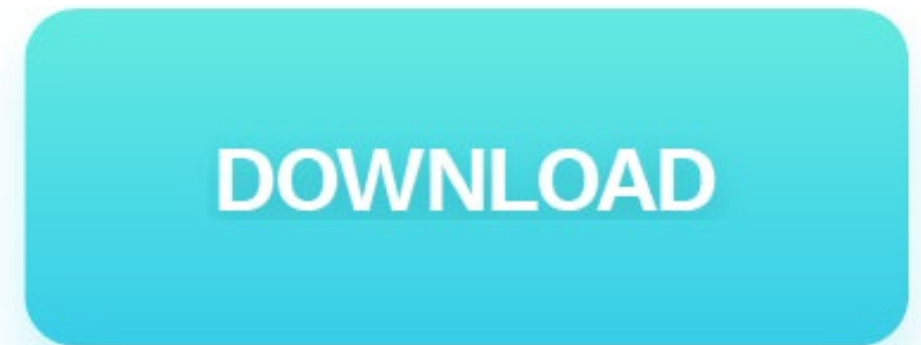
Gradients

Gradients help to achieve the Skeuomorphism look of 3d, realistic buttons.



Classic Vs Modern Gradients!

Modern gradient, without any extra decoration, can work with a soft shadow made from the same color, creating a beautiful, semi-realistic look.



WHY DO WE LOVE GRADIENTS?



THE POWER OF GRADIENTS

Depth and a more
defined shape.

Catching the eye
and guiding it to
certain parts

FLAT BUTTON

GRADIENT

GRADIENT + SHADOW

GRADIENTS IN BRANDING



Tickety



datefinder

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GRADIENT TYPES (LINEAR)



Horizontal, Two Colors.
Left to right.



Vertical, Two colors.
Top to bottom.



Diagonal, Two colors.
45 degree angle.



Horizontal, Three Colors.
Left to right.



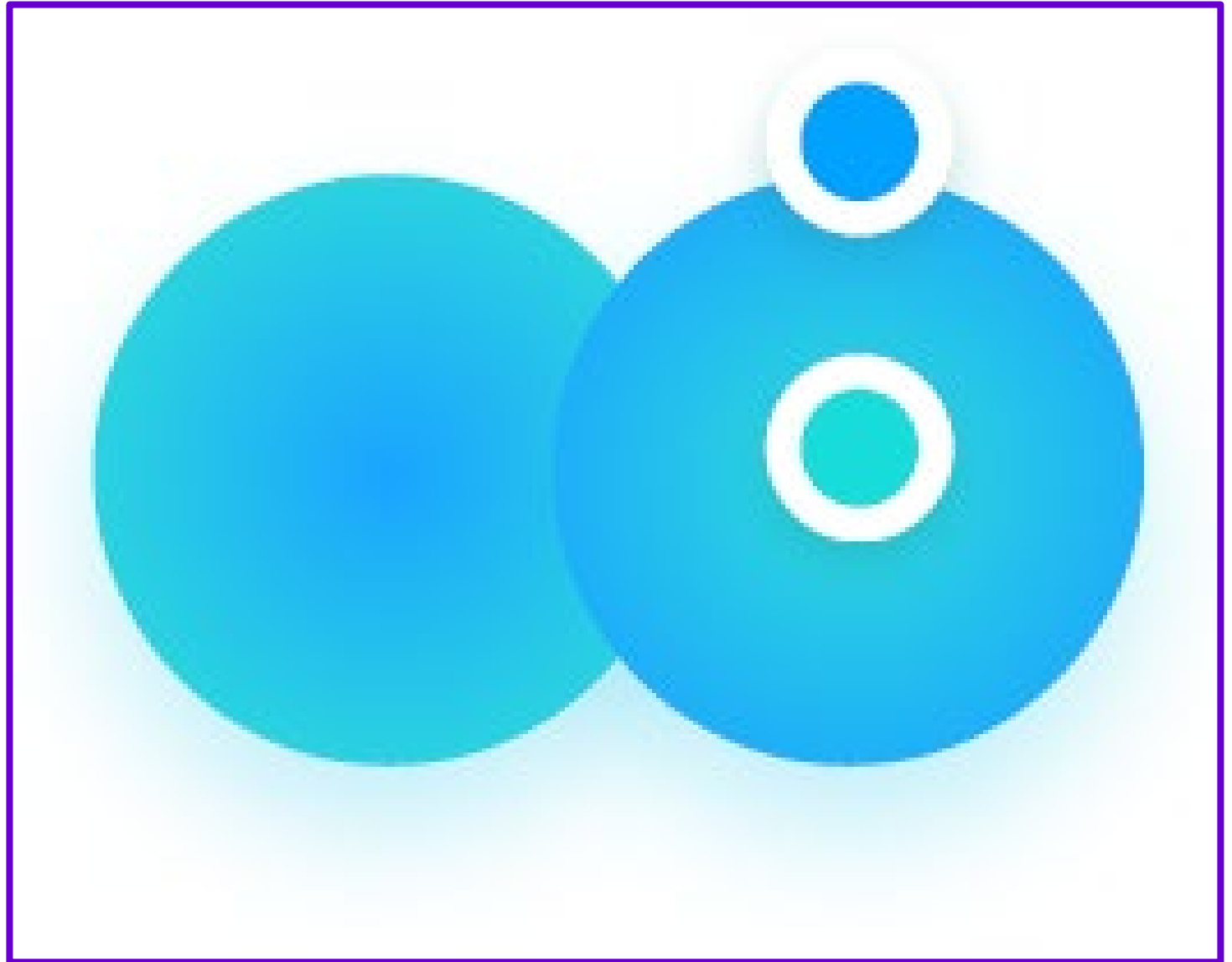
Vertical, Three Colors.
Top to Bottom.



Diagonal, Three Colors
45 degree angle.

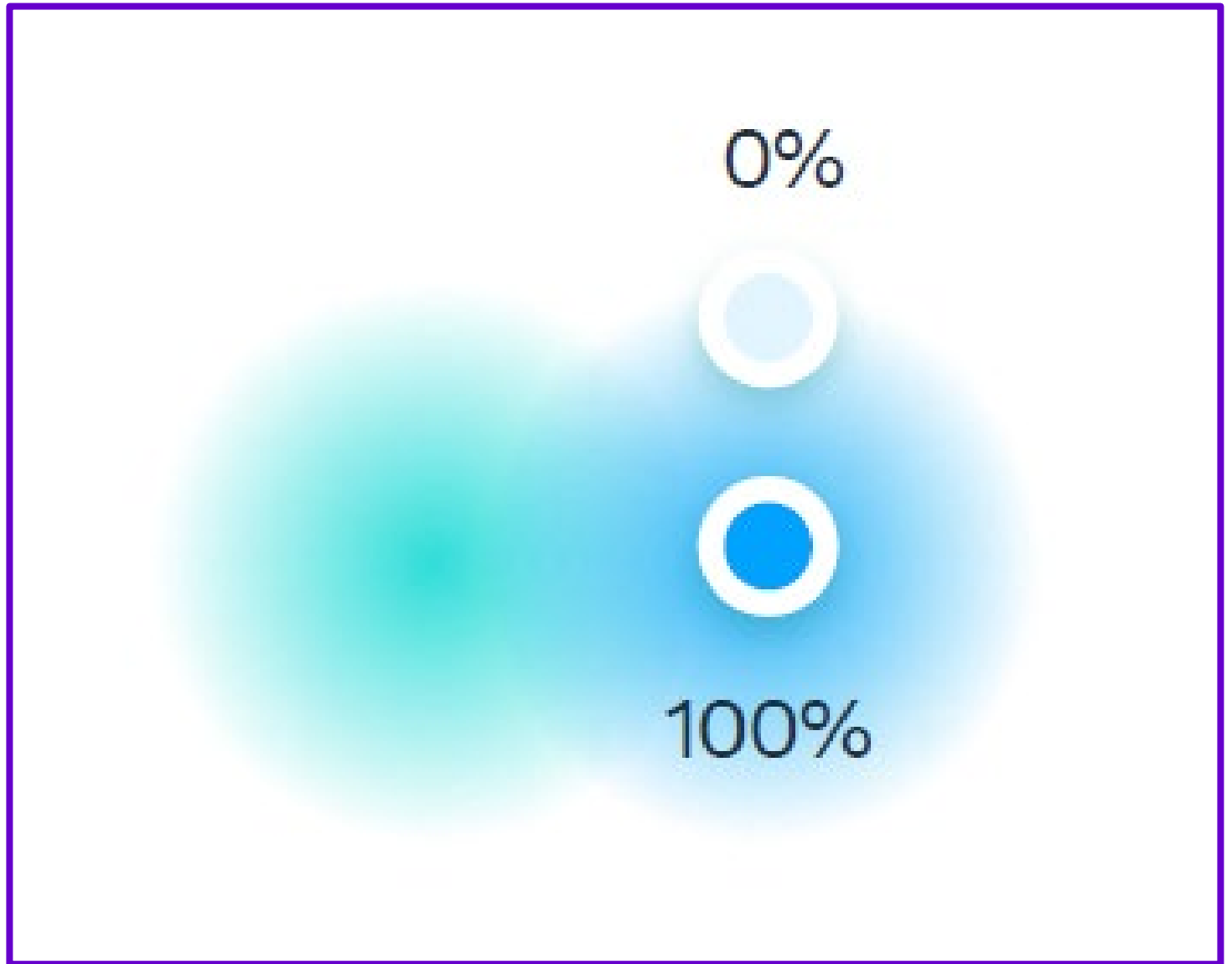
GRADIENT TYPES (RADIAL)

Two colors with no transparency can create a 3d look in circles by adding a highlight and shading.



GRADIENT TYPES (RADIAL)

If both ends of the gradient are the same color, with one at full transparency, we end up with a blur or fade-out effect.



GRADIENT TYPES (RADIAL)

A radial gradient works well on non-circular shapes by adding a bit of an organic, realistic style to them.



GRADIENT TYPES (ANGULAR)

Two colors set in the same spot will have a sharp edge between them, while the gradient goes around the other way.



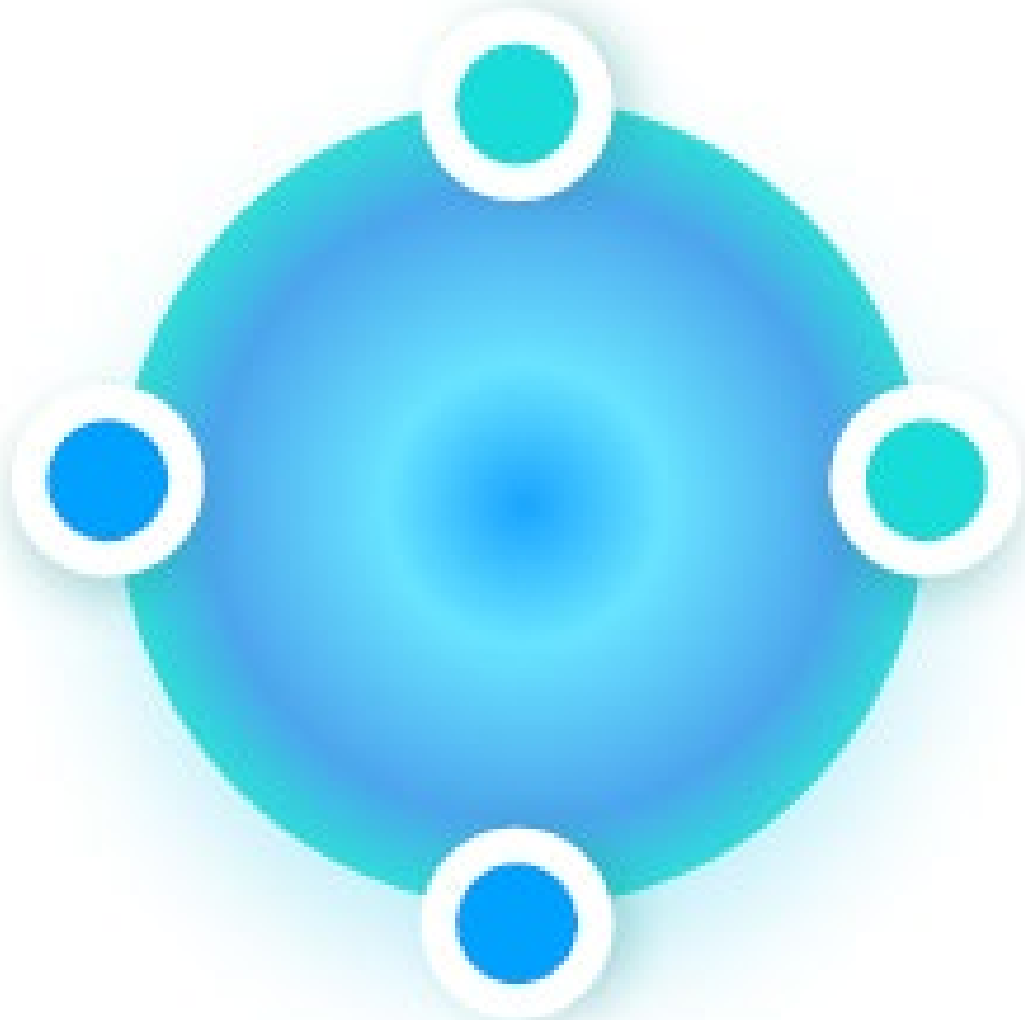
GRADIENT TYPES (ANGULAR)

Two colors at 180-degrees will have a fluid transition between the colors going both ways.



GRADIENT TYPES (ANGULAR)

You can add more colors and play around with the angle to achieve some interesting results.



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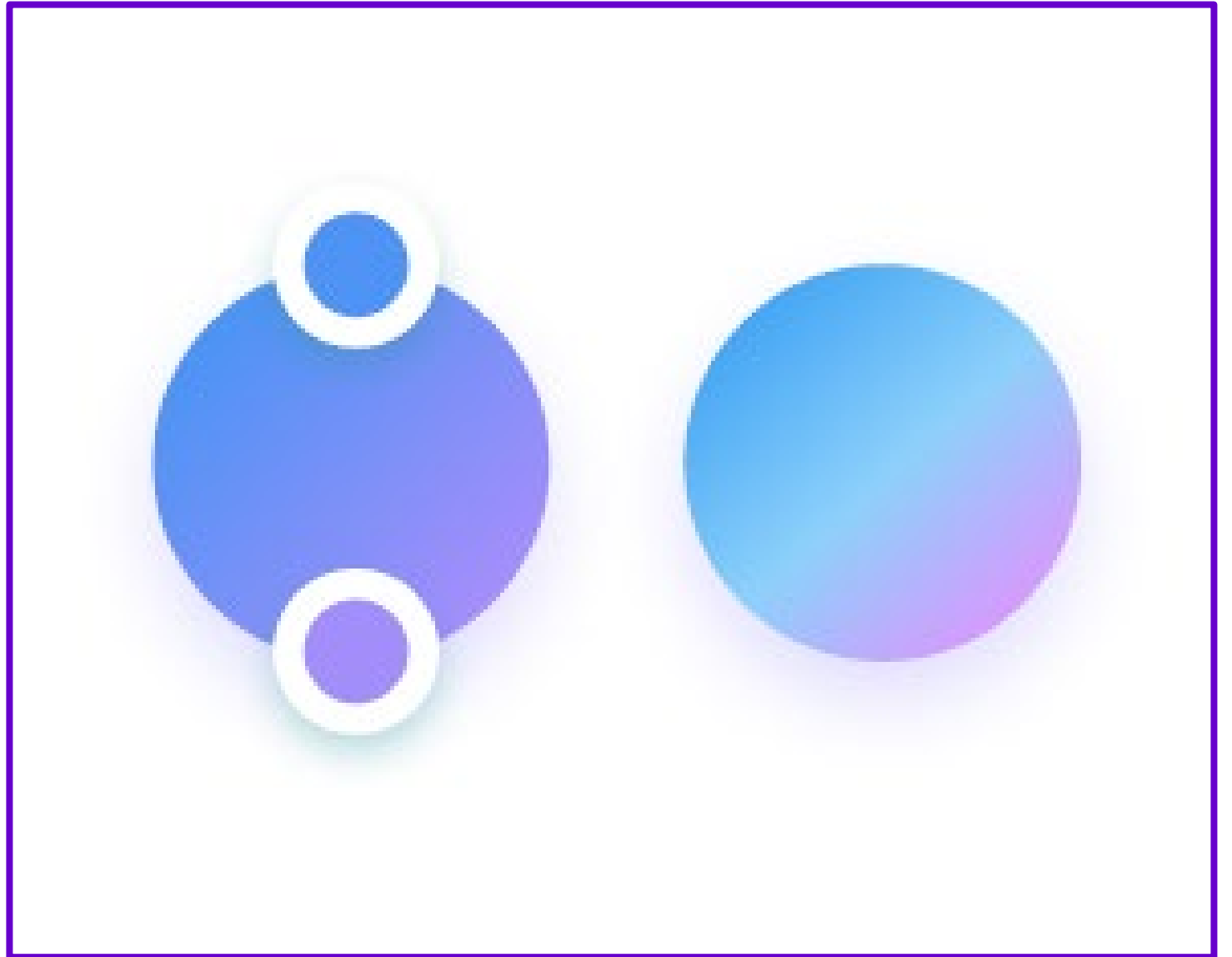
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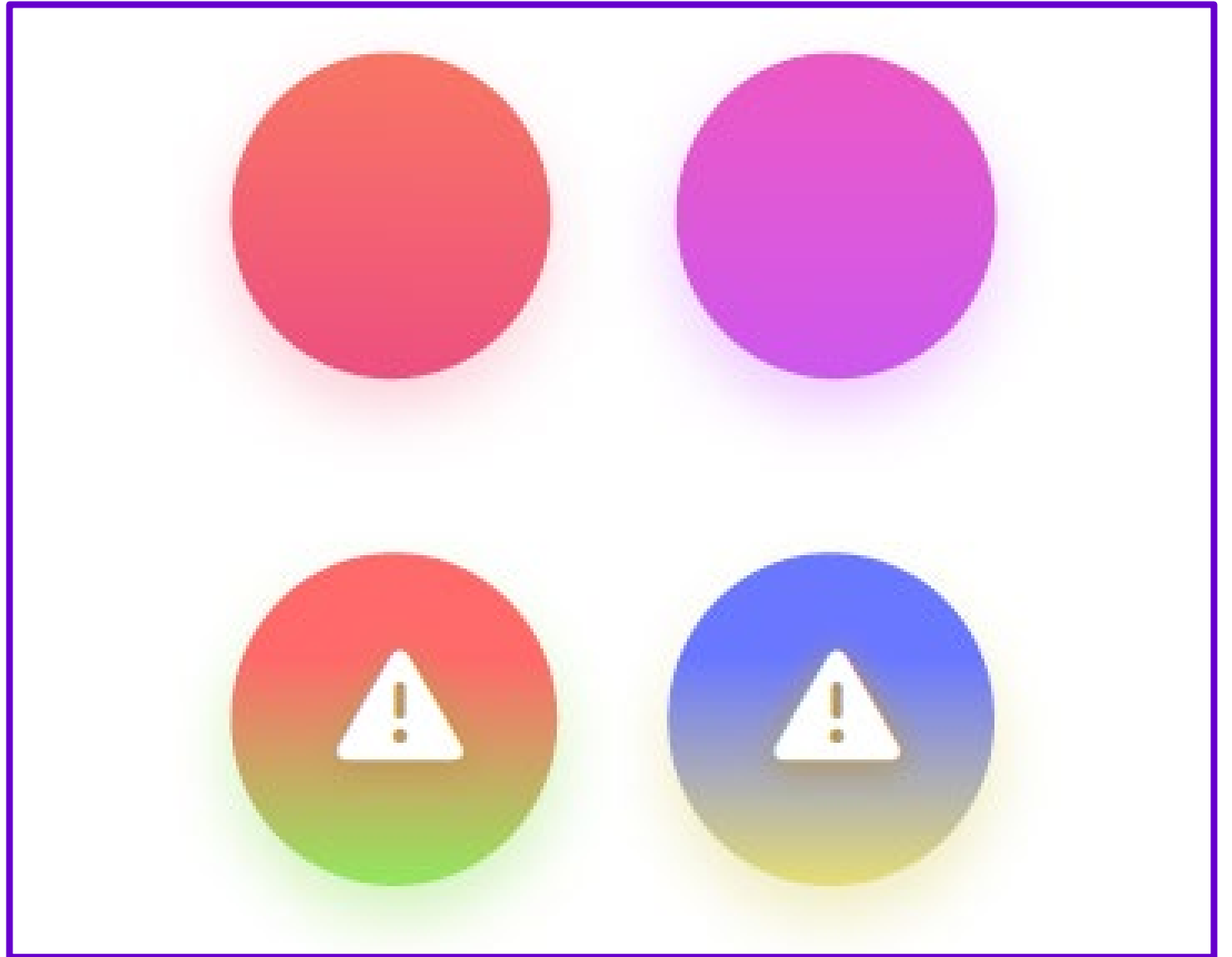
BEST PRACTICES (TWO COLORS)

Two-color gradients are the best choice for most designs. With more colors, the gradient can become visually busy.



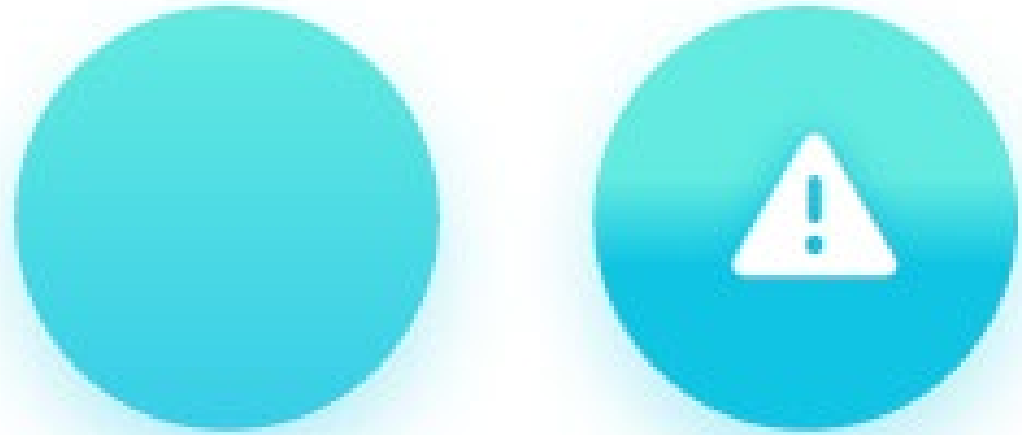
BEST PRACTICES (RIGHT COLORS)

Try to match similar colors, warm hues with warm ones, and cold with cold. That will help you avoid bad mixes



BEST PRACTICES (SUBTLE GRADIENTS)

The best gradients are often very subtle, with not a lot of difference between the colors. It makes them look more natural and easier on the eyes.



BEST PRACTICES (VERTICAL)

Vertical is the "classic" button gradient. If the lighter color is at the top, it looks more 3-dimensional and almost asking the user to activate it



PUSH ME



I'M PRESSED

BEST PRACTICES (DIAGONAL)

Diagonal gradients help us achieve that organic, natural look. If you're trying them out for the first time, start safe, with a 45-degree angle



PUSH ME

BEST PRACTICES (DIAGONAL)

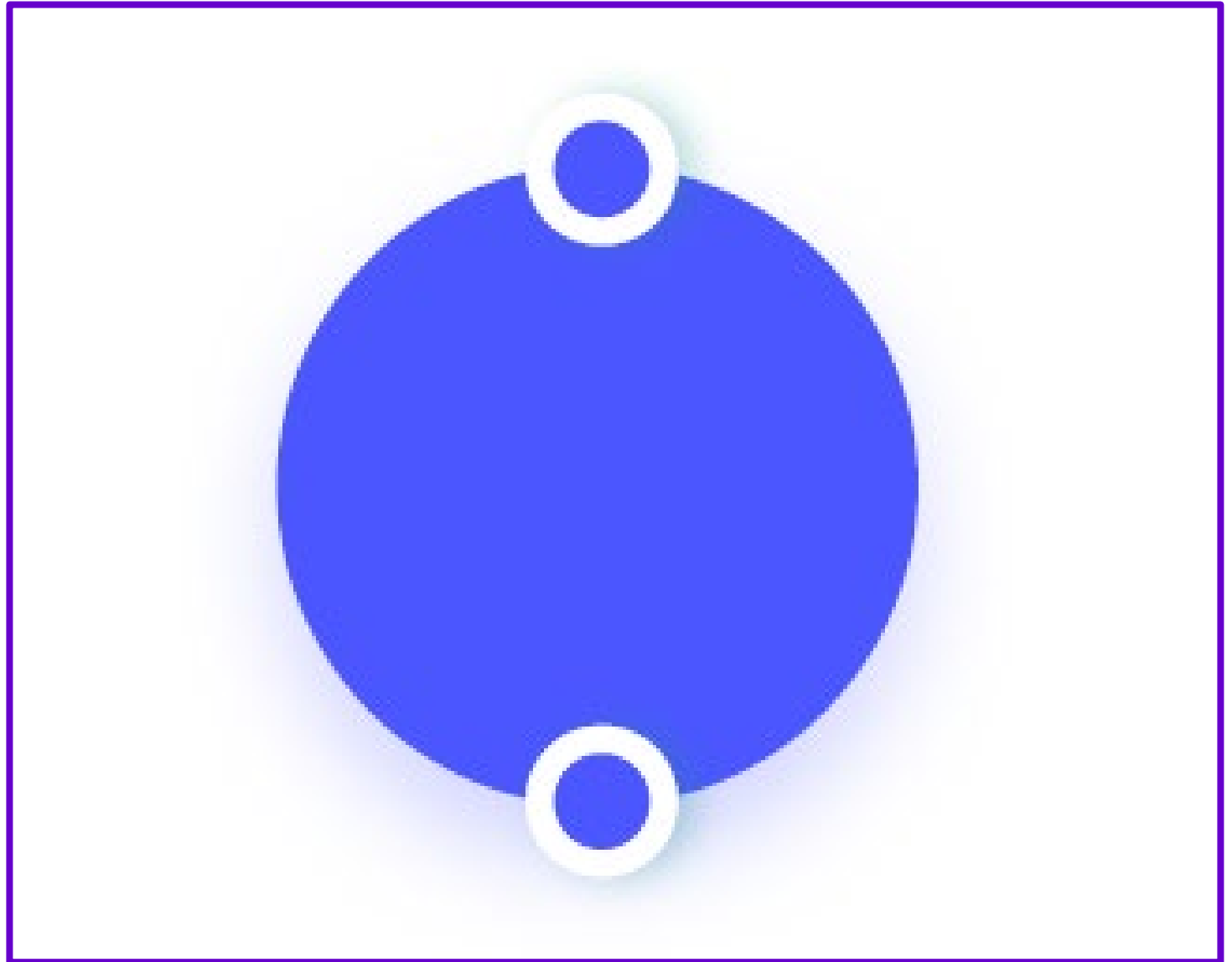
Diagonal gradients help us achieve that organic, natural look. If you're trying them out for the first time, start safe, with a 45-degree angle



PUSH ME

How TO CREATE GRADIENTS (STEP 1)

Create a linear gradient and use the same color on both ends. In our case, it's simply a transition from #4B56FF to #4B56FF.



How TO CREATE GRADIENTS (STEP 2)

Check if your color palette is in HSB mode (Hue, Saturation, Brightness) and decrease or increase the Hue value of one of the gradient endings by 15-30.

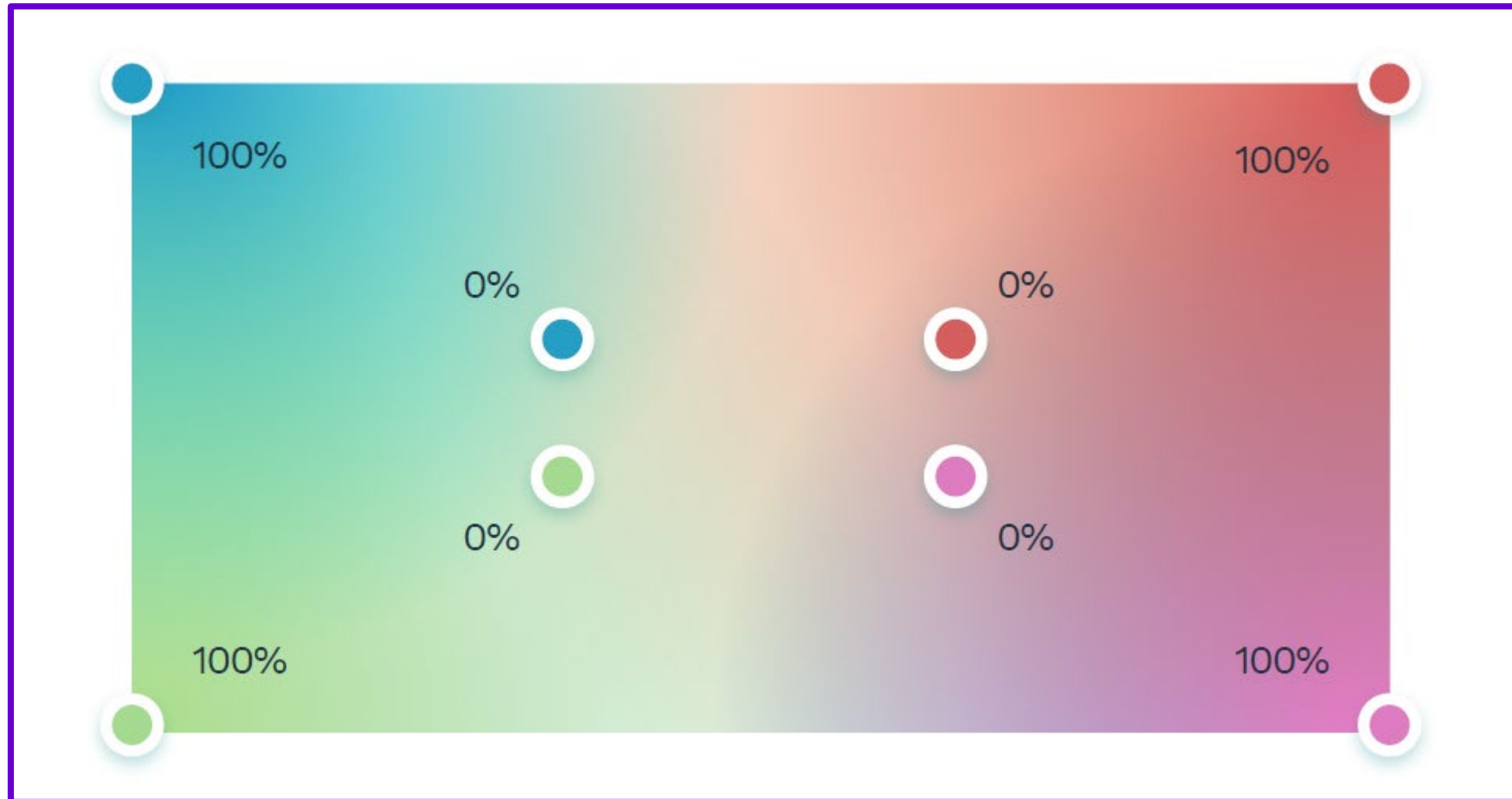


How TO CREATE GRADIENTS (STEP 3)

For a more organic, friendly effect, you can try rotating the gradient (here it's at 45 degrees) or decreasing the Saturation by 10-15 points.



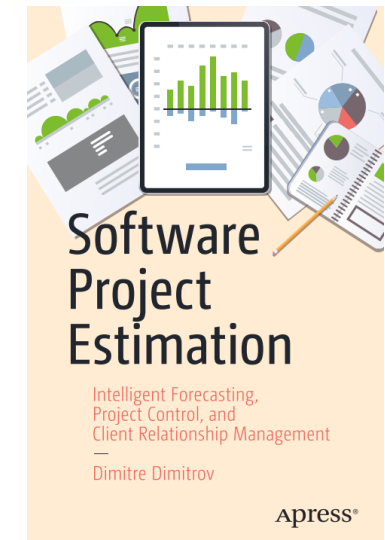
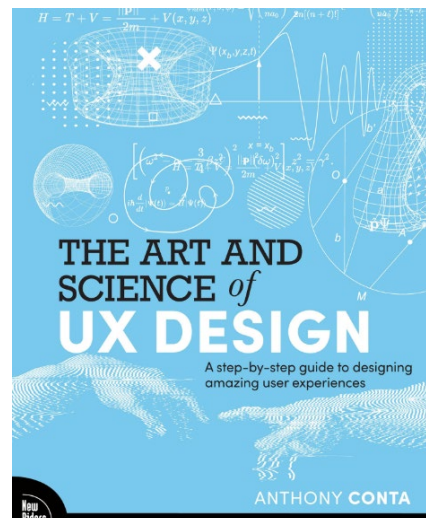
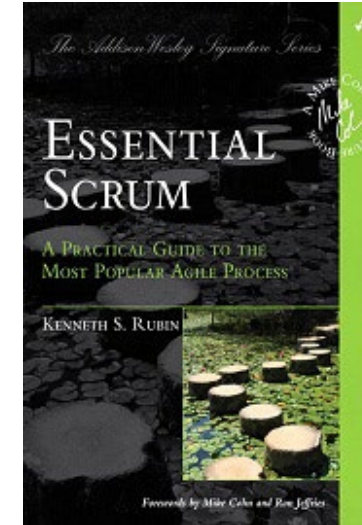
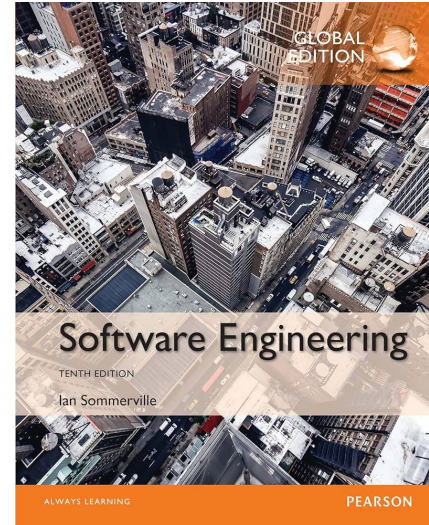
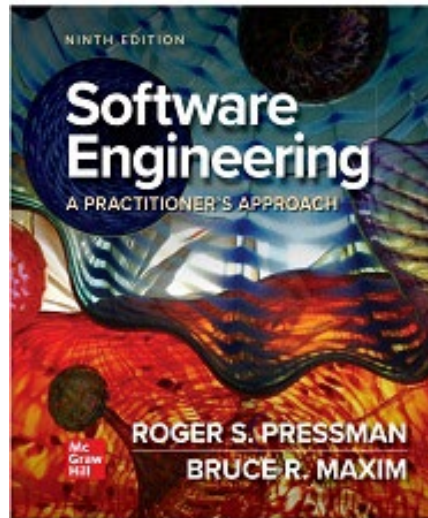
SOFT GRADIENT



SUBTLE GRADIENTS

Subtle gradients between similar colors are perceived as the most beautiful.

Course References



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